Trajecta: A Heterogeneous, Traffic-Aware and Infrastructure-Conscious Smart Logistics Optimization Framework

Gopika Rani N. [a,1], Dinesh T M [a], Harish Karthic R [a], Madhan KJ [a] and Rahul R [a]

[a] *Department of Computer Science and Engineering, PSG College of Technology, Coimbatore, India*

Emails: {ngr.cse, 23z320, 23z324, 23z340, 24z465}@psgtech.ac.in

Abstract. Routing optimization for logistics networks has grown considerably harder as fleets diversify and cost structures extend beyond simple distance. This paper presents **Trajecta**, an extension of the earlier *Profecta* system that combines heterogeneous fleet modeling, three-dimensional (3D) capacity validation, traffic-aware travel-time estimation, infrastructure feasibility checking, stochastic service-time modeling, and a seven-factor cost model within a single heuristic pipeline. Eight order-splitting strategies and seven routing heuristics generate 56 candidate routes per seed, from which three routes (minimum cost, minimum distance, minimum penalty) are selected independently. On a synthetic 30-order, 18-city instance, with all methods scored on the same seven-factor cost, Trajecta's minimum-cost route is on average 5.8% cheaper, 5.9% shorter and has 32.5% lower penalty than OR-Tools CP-SAT, with +0.48 percentage points on-time delivery (mean over 10 seeds). The difference is small compared to the standard deviation, and no significance test was done. An ablation shows that removing DBSCAN splitting clearly hurts (cost +22%, distance +31%), while the other modules change results little on these instances. On five public benchmark instances the outcome against OR-Tools is mixed (one win, one tie, three losses). The main experiment is synthetic and the infrastructure data are assumed.

Keywords. Routing Optimization, Heuristic Search, Mixed Fleets, Stochastic Model, Vehicle Routing Problem

1. Introduction

E-commerce growth has pushed last-mile and inter-city transportation costs to the forefront of logistics planning [1]. Fuel, the operational cost of the transport mode, driver salary, late-delivery penalties, premium service cost, and infrastructure cost [1] make up a substantial part of logistics cost. Existing routing algorithms

[1]Corresponding Author: Gopika Rani N., Department of Computer Science and Engineering, PSG College of Technology, Coimbatore, India; E-mail: ngr.cse@psgtech.ac.in.

usually make the best route on the basis of distance or time of travel, under the assumption that all the vehicles are identical and traffic remains consistent [2].

But in the current case study, the vehicle fleet could be of any type ranging from two-wheelers to multi-axle diesel trucks [3]. Delays are time-varying, weather can add significant time to a trip segment, and bridge or road-weight restrictions can rule out a route for a given vehicle entirely [4,5]. Off-the-shelf optimization models generally ignore all of this.

This paper introduces **Trajecta**, an extension of *Profecta* [6] that integrates heterogeneous fleet management, 3D capacity checking, traffic-aware routing, infrastructure feasibility checking, and a seven-factor cost model. Trajecta targets lower energy cost through vehicle–order matching and fewer late-delivery penalties through time-sensitive prioritization; workload balance is not evaluated here. A geospatial visualization layer lets planners inspect the selected routes before dispatch.

Contributions: (i) 3D capacity checking prior to vehicle assignment; (ii) multiplicative, multi-factor travel-time modeling; (iii) infrastructure feasibility checks integrated before route scoring; (iv) stochastic service-time estimation; and (v) an evaluation against OR-Tools CP-SAT [15], ALNS, a Genetic Algorithm (GA) and Tabu Search under one shared seven-factor cost, with an ablation, a limited scalability test and five public benchmark instances. Demonstrated results are separated from future work in Sections 5–7.

2. Related Work

VRP research has moved from pure distance minimization toward more realistic formulations: learning-based routing [2]; cost-oriented models combining freight, warehousing, and carbon-budget objectives [7,8]; and metaheuristics—ACO, PSO, and hybrid ACO–Tabu Search—reported to outperform exact solvers under time constraints [9,10]. Split-delivery VRPs have established complexity benchmarks for order splitting [11], and 3D-loading-constrained formulations have addressed volumetric packing [12,13]. Heterogeneous-fleet heuristics with time windows and priorities [3] and traffic-aware or real-time routing [4,5] address mixed fleets and congestion, respectively, but separately from 3D loading, infrastructure restrictions and a multi-term cost model.

Gap: Among the works cited here, none combines a heterogeneous fleet with 3D capacity checking, time-varying traffic, infrastructure feasibility checks, stochastic service times and a seven-factor cost model; Trajecta addresses this combination. Table 1 summarizes how Trajecta extends Profecta [6].

3. Methodology

3.1. Problem Setup and Data

The main experiment is synthetic. It uses 30 orders over 18 named city entries around Coimbatore (including the depot). Order weights are 4–55 kg, volumes

Table 1. Profecta [6] versus Trajecta

Aspect	Profecta	Trajecta
Fleet	Uniform fleet	Heterogeneous, 7 vehicle types
Cost factors	5	7 (adds tolls and EV charging)
Routing heuristics	3	7
Order-splitting strategies	6	8 (adds Penalty-Risk and DBSCAN)
Additional checks and models	—	3D capacity check, infrastructure feasibility check, traffic-aware times, stochastic service times

0.015–0.24 m^3, due-time offsets 4–8 h and arrival times 6–10 h; 15% of orders are fragile and 20% need cooling. Distances are haversine distances multiplied by a road factor of 1.18. Coordinates and infrastructure constraints are generated by the implementation; the infrastructure constraints are synthetic assumptions based on city-pair rules, not real infrastructure data. The 3D capacity check greedily places boxes into the available free spaces, trying axis permutations. The fleet has seven vehicle types (petrol/electric two-wheelers, petrol/diesel/electric four-wheelers, diesel/electric trucks; full specification in Supplement S1) and four static traffic bands (morning 1.30, evening 1.25, midday 1.05, night 0.90). The weather and lane factors are both 1.0. Generation details are in Supplement S2.

3.2. System Architecture

Trajecta converts an unordered order list into ordered routes in six steps (Figure 1): splitting, vehicle allocation, route creation, cost calculation, route selection with visualization, and dispatcher review (Sections 3.3–3.8).

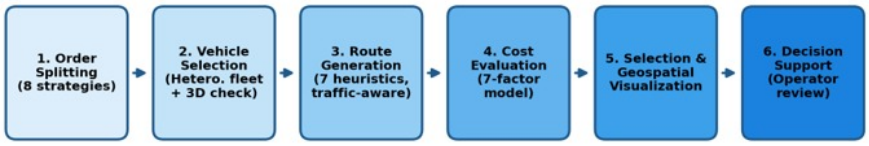


Figure 1. System architecture and flow of Trajecta

3.3. Order Splitting and Routing Heuristics

Orders are split by Estimated Delivery Date, Arrival Date, Latitude, Travel Time, Longitude, combined Latitude–Longitude, Penalty-Risk, and Density-Based (DBSCAN) clustering [16]; the last two are new relative to the six strategies inherited from Profecta [6]. Within each split, routes are generated by seven heuristics—Nearest Neighbor, Bi-Directional Search, Most Urgent Destination (EDD-ordered), Farthest Insertion, Randomized Nearest Neighbor, Greedy Time-Window, and Cluster-First Route-Second—all starting and ending at the depot. This gives $8 \times 7 = 56$ candidate routes per seed.

3.4. Traffic-Aware Travel Time

$$T_{ij}^{v,t} = \frac{d_{ij}}{s_v} \times \lambda_v \times \omega \times \tau_t \tag{1}$$

where d_{ij} is the segment distance, s_v the vehicle base speed, λ_v the lane factor, ω the weather multiplier and τ_t the time-of-day multiplier (Table 2). Factors compound multiplicatively, so several adverse conditions degrade travel time faster than their sum. In our runs $\lambda_v = \omega = 1.0$, so only the four time-of-day bands change travel times.

3.5. Stochastic Service Time

$$S_i = t_{\text{base}} + \alpha w_i + \beta V_i + \gamma(4f_i + 6c_i) + \delta\varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \tag{2}$$

Service time at each stop depends on order weight w_i, volume V_i, fragility and cooling flags f_i, c_i (+4 and +6 min), and zero-mean Gaussian noise ε_i with $\sigma =$ 2 min. The values are $t_{\text{base}} = 8$ min, $\alpha = 0.05$ min/kg, $\beta = 5$ min/m^3 and $\gamma = \delta = 1.0$. They are modeling assumptions in the spirit of aggregate service-time statistics in last-mile literature [14], not fitted to real transaction data.

3.6. Seven-Factor Cost Model

$$\text{TotalCost(route)} = EC + WF + PC + PK + MC + TC + CC \tag{3}$$

where EC is energy cost (distance × per-km rate: petrol Rs. 4.5, diesel Rs. 7, electric Rs. 1.1), WF is wages (Rs. 2/km) plus the vehicle's fixed cost, PC is the late penalty (Rs. 100 per late day, summed over orders), PK is packaging (Rs. 35 per flagged order), MC is maintenance (Rs. 0.55/km), TC is tolls (Rs. 30 per route) and CC is EV charging (Rs. 0.35/km); a cost scale factor of 1.16 is part of the configuration (Supplement S1, S4). This extends Profecta's five-factor model [6] with tolls and EV charging.

3.7. Mathematical Formulation

Define O to be the set of orders (where $|O| = 30$), V to be the set of vehicles, and C to be the set of cities with the depot ($|C| = 18$). Each order o has weight w_o, volume V_o and a due time D_o. The decisions are the batch-to-vehicle assignment $x_{ov} \in \{0, 1\}$ and, for each vehicle v, the visit order π_v of the cities it serves, starting and ending at the depot. With $B_v = \{o : x_{ov} = 1\}$:

$$\min \sum_{v \in V} \text{TotalCost}(\pi_v, B_v) \quad \text{s.t.} \quad \sum_v x_{ov} = 1 \ \ \forall o \tag{4}$$

$$\sum_{o \in B_v} w_o \le W_v, \quad \sum_{o \in B_v} V_o \le Q_v, \quad \text{Pack3D}(B_v, v) = \text{feasible}, \quad \text{Infra}(\pi_v, v) = \text{feasible} \tag{5}$$

Due times are handled by penalty rather than as hard constraints: arrival times follow from Eqs. (1)–(2) and a late order adds to PC in Eq. (3). The full formulation is in Supplement S3.

3.8. Candidate Selection and Complexity

The 56 candidates per seed are scored with Eq. (3). Three routes are then selected independently by arg min over the 56 candidates: minimum total cost, minimum distance and minimum penalty. No scalarization or weighted combination is used, so the trade-offs between the three stay visible to the dispatcher. The dominant cost is the $\mathcal{O}(N^2)$ heuristic evaluation repeated $S \times H = 56$ times, giving $\mathcal{O}(SH(N^2+E+N))$, where E is the number of infrastructure checkpoints.

4. Experimental Setup

Baselines. On the synthetic instance Trajecta is compared with OR-Tools CP-SAT [15], ALNS, GA and Tabu Search, all scored with the same seven-factor cost. Trajecta and CP-SAT both use a 10 s time limit. CP-SAT is the only baseline reported with full detail here; ALNS, GA and Tabu Search are summarized in Supplement S6. **Metrics.** Cost, distance and penalty are reported as mean ± SD over seeds. On-time % is the share of orders arriving by their due time. No statistical significance test was run, so results are described only as consistent or inconsistent in direction. The public-benchmark comparison uses a different objective (vehicle count first, distance second), not the seven-factor cost [17,18]. Table 2 lists the setup; code, data and result files are at `https://github.com/tmdinesh/Trajecta` (Supplement S9).

5. Results and Discussion

5.1. Synthetic Comparison

Table 3 compares Trajecta's minimum-cost candidate with OR-Tools CP-SAT. Under the strict 10 s limit CP-SAT struggled to converge: it averaged a 90.31% optimality gap and returned no feasible solution on 1 of 10 seeds, so its statistics use 9 feasible runs. Trajecta was feasible on all seeds. They are consistent in sign but relatively small compared to standard deviations; they are inconclusive, and no statistical test has been done yet.

Against ALNS, GA and Tabu Search, Trajecta's best-candidate means are lower in cost and distance. However, ALNS and Tabu Search have a lower mean penalty (Rs. 23.20, versus Rs. 34.80 for Trajecta's minimum-cost route) and slightly higher on-time delivery (99.33% versus 99.00%), so Trajecta does not dominate them on penalty. Trajecta's runtime was not recorded, so no speed claim is made.

Table 2. Experimental setup and key parameters

Item	Value
Software	Python 3.12.9, OR-Tools 9.15.6755, NumPy 2.3.5, pandas 3.0.1, SciPy 1.18.1, Matplotlib 3.11.2
Hardware	AMD Ryzen 5 5600H, 15.4 GB RAM
Synthetic runs	Seeds 0–9; 30 orders; 18 cities incl. depot; time limit 10 s (Trajecta and OR-Tools)
Benchmark runs	Seeds 11, 22, 33 per instance; time limit 2 s; Solomon c101, r101, rc101; Cordeau MDVRP p01, p02
Ablation / scalability	3 seeds per ablation variant; 2 seeds per scalability size (20, 30, 50, 80 orders)
Traffic multipliers	Morning 1.30, evening 1.25, midday 1.05, night 0.90; weather and lane factors 1.0
Service time	Base 8 min; 0.05 min/kg; 5 min/m^3; +4 min fragile; +6 min cooling; noise SD 2 min; $\gamma = \delta = 1.0$
Cost rates (Rs.)	Petrol 4.5/km; diesel 7/km; electric 1.1/km; wages 2/km; maintenance 0.55/km; late penalty 100/day; packaging 35 per flagged order; toll 30/route; EV charging 0.35/km; scale factor 1.16
Fleet	7 types, 35 kg / 0.18 m^3 (2W-Petrol) up to 4000 kg / 25 m^3 (Truck-Diesel); full table in Supplement S1

Table 3. Synthetic comparison, mean ± SD (10 seeds; OR-Tools over its 9 feasible runs)

Metric	Trajecta (min-cost)	OR-Tools CP-SAT	Trajecta vs CP-SAT
Cost (Rs.)	$4,278.20 \pm 304.03$	$4,539.50 \pm 367.78$	5.8% lower
Distance (km)	470.61 ± 32.34	500.17 ± 41.31	5.9% lower
Penalty (Rs.)	34.80 ± 56.03	51.56 ± 61.14	32.5% lower
On-time (%)	99.00 ± 1.61	98.52 ± 1.76	+0.48 pp
Feasible runs	10 of 10	9 of 10	—

5.2. The Three Selected Routes

Table 4. Independently selected candidates, mean ± SD over 10 seeds

Selection	Cost (Rs.)	Distance (km)	Penalty (Rs.)	On-time (%)
Min cost	$4,278.20 \pm 304.03$	470.61 ± 32.34	34.80 ± 56.03	99.00 ± 1.61
Min distance	$4,293.24 \pm 310.24$	466.82 ± 29.75	69.60 ± 81.11	98.00 ± 2.33
Min penalty	$5,451.42 \pm 1,124.65$	633.04 ± 149.03	0 ± 0	100 ± 0

Table 4 shows the real trade-off. Eliminating penalty entirely costs on average about 27% more in total cost and about 35% more distance than the minimum-cost selection, and its cost varies much more across seeds. Minimum-cost and minimum-distance selections are close in cost but the latter carries about twice the mean penalty. All 56 candidates per seed and the per-seed selections are in the repository (Supplement S5).

5.3. Ablation

Table 5 removes one module at a time (3 seeds per variant; the baseline includes DBSCAN splitting). The baseline cost (Rs. 3,568.80) differs from the 10-seed mean in Table 3 because it uses a different seed set.

Table 5. Ablation, mean over 3 seeds per variant

Variant	Mean cost (Rs.)	Mean distance (km)	Mean on-time (%)
Baseline	3,568.80	363.08	100.00
No 3D packing	3,568.80	363.08	100.00
No infrastructure	3,550.94	363.08	100.00
No DBSCAN	4,362.75	474.12	97.78
No penalty-risk split	3,572.84	363.87	100.00
No traffic	3,547.46	358.67	100.00
No stochastic service time	3,558.40	358.67	100.00

Removing DBSCAN clearly hurts (cost +22%, distance +31%, on-time 97.78%). The other modules change results little or not at all on these instances. The 3D check gave identical results, so its effect is not visible at this scale or on these instances. Removing infrastructure, traffic or service-time modeling slightly lowers cost, as expected, because the constraints and added delays are removed. These modules are therefore justified by feasibility and realism, not by cost reduction on this data.

5.4. Scalability

Scalability is limited. With 2 seeds per size, OR-Tools returned scored runs only at 20 orders (Rs. 1,994.16, 212.18 km) and 30 orders (Rs. $4,548.46 \pm 9.59$, 492.09 ± 6.41 km); the saved 50- and 80-order runs returned no feasible score, and the 80-order metaheuristic scores are recorded as infeasible. Mean OR-Tools runtimes were about 10.03, 10.07, 10.11 and 11.32 s at 20, 30, 50 and 80 orders. Scaling beyond 30 orders was therefore not demonstrated for the baselines, and no Trajecta results are reported for these sizes (Supplement S7).

5.5. Public Benchmarks

Public benchmarks use a vehicle-count-first, distance-second objective, not the seven-factor cost, with 3 seeds per instance (Table 6). All methods were feasible in 15 of 15 runs. Trajecta wins 1 instance, ties 1 and loses 3; total vehicle counts tie across the set and OR-Tools has the lower aggregate distance. These instances check general routing competence, not the heterogeneous, 3D or infrastructure features, and the outcome is mixed [17,18].

Table 6. Benchmark results, mean (vehicles, distance) over 3 seeds

Instance	Trajecta	OR-Tools	Winner
Cordeau p01	12, 613.728	12, 609.241	OR-Tools
Cordeau p02	7, 517.315	7, 507.011	OR-Tools
Solomon c101	10, 828.937	10, 828.937	Tie
Solomon r101	23, 1,994.482	22, 1,789.111	OR-Tools
Solomon rc101	17, 1,976.042	18, 1,806.160	Trajecta

6. Limitations

Orders are synthetic and infrastructure rules are generated assumptions; no real fleet or traffic data were used. No significance test was run. CP-SAT was not proven optimal (mean gap 90.31%; 9 feasible and 1 unknown of 10 seeds), so no optimality conclusions are drawn. Scaling beyond 30 orders was not successfully demonstrated for the baselines. The ablation uses only 3 seeds and shows no visible effect of the 3D check. Traffic uses four static bands, Trajecta's runtime was not recorded, and EV charging assumes a static tariff approximated at Rs. 0.35/km. Routes are computed offline with no replanning after dispatch.

7. Conclusion and Future Work

Trajecta extends Profecta with 3D capacity checking, multiplicative traffic-aware travel times, infrastructure feasibility checking and stochastic service times under a seven-factor cost. **Demonstrated:** on a synthetic 30-order instance and under a shared seven-factor cost, its minimum-cost route was on average 5.8% cheaper and 5.9% shorter than OR-Tools CP-SAT over 10 seeds, with modest, directionally consistent differences in penalty and on-time delivery; DBSCAN splitting was the one module with a clear effect; benchmark results were mixed. **Not demonstrated:** speed, optimality, scalability beyond 30 orders, and real-world performance. Future work includes recording and comparing runtimes, live-traffic integration, validation with an industry partner, multi-depot extensions, learned traffic multipliers, and a significance analysis over more seeds.

Declaration of Generative-AI Use

Generative AI tools were used to generate the code and the synthetic data used to run the experiments. They were not used for any other purpose.

Data and Code Availability

Code, synthetic orders (`synthetic_30_orders.csv`) and result files: `https://github.com/tmdinesh/Trajecta`. Supplementary material accompanies this paper.

References

[1] Y. Zhao, Y. Zhou and W. Deng, Innovation mode and optimization strategy of B2C e-commerce logistics distribution under big data, *Sustainability* **12** (2020), 3381.

[2] A. Snoeck, D. Merchan and M. Winkenbach, Route learning: a machine learning based approach to infer constrained customers in delivery routes, *Transportation Research Procedia* **46** (2019), 229–236.

[3] M. A. Mohammed, N. A. Saleh and D. J. Hoose, Heuristic algorithms for the heterogeneous vehicle routing problem with time windows, customers priority, pickup and delivery, *IEEE Access* **13** (2025), 67820–67845.

[4] P. Kumar, S. Sharma and A. Patel, Real time routing with vehicle failure and traffic awareness in last-mile delivery, *Engineering Applications of Science and Letters* **3** (2026), 15–28.

[5] P. Alvarez, L. Lerga, A. Serrano-Hernandez and J. Faulin, The impact of traffic congestion when optimizing delivery routes in real-time: a case study in Spain, *International Journal of Logistics Research and Applications* **22**(3) (2019), 219–234.

[6] G. Rani and Swetha, Profecta: A smart logistics system, in: *Data Science & Exploration in Artificial Intelligence (CODE-AI 2024)*, A A Balkema, 2025.

[7] G. Qin, F. Tao and L. Li, A vehicle routing optimization problem for cold chain logistics considering customer satisfaction and carbon emissions, *International Journal of Environment Research and Public Health* **16** (2019), 576.

[8] E. Sert, L. Hedayatifar, R. A. Rigg, A. Akhavan, O. Buchel, et al., Freight time and cost optimization in complex logistics networks, *Complexity* (2020).

[9] W. Liu, Route optimization for last-mile distribution of rural e-commerce logistics based on ant colony optimization, *IEEE Access* **8** (2020), 12179–12187.

[10] A. Sifaleras and I. Konstantaras, Variable neighborhood descent heuristic for solving reverse logistics multi-item dynamic lot-sizing problems, *Computers & Operations Research* **78** (2015), 385–392.

[11] D. Gulczynski, B. Golden and E. Wasil, The split delivery vehicle routing problem with minimum delivery amounts, *Transportation Research Part E* **46**(5) (2010), 612–626.

[12] Y. Chen, X. Liu and W. Zhang, A study of the capacitated vehicle routing problem with time-window and three-dimensional loading constraints in land–sea transport, *Sustainability* **12** (2024), 4523.

[13] H. Wang, M. Zhang and Q. Liu, An improved approach for vehicle routing problem with three-dimensional loading constraints based on genetic algorithm and residual space optimized strategy, *Processes* **13** (2025), 1487.

[14] A. K. Singh, R. Verma and P. Sharma, The vehicle routing problem with simultaneous pick-up and delivery under fuzziness considering fuel consumption, *Logistics* **8**(2) (2024), 42.

[15] L. Perron and V. Furnon, OR-Tools, Google, `https://developers.google.com/optimization`.

[16] M. Ester, H.-P. Kriegel, J. Sander and X. Xu, A density-based algorithm for discovering clusters in large spatial databases with noise, *Proceedings of KDD-96* (1996), 226–231.

[17] M. M. Solomon, Algorithms for the vehicle routing and scheduling problems with time window constraints, *Operations Research* **35**(2) (1987), 254–265.

[18] J.-F. Cordeau, M. Gendreau and G. Laporte, A tabu search heuristic for periodic and multi-depot vehicle routing problems, *Networks* **30**(2) (1997), 105–119.